

The Impact of Visually Grounded Descriptions on Acceptance of Robot Obstacle-Handling Behaviors

Sure Bak
NAVER LABS

Seongnam-si, Republic of Korea
sure.bak@naverlabs.com

Seijin Cha
NAVER LABS

Seongnam-si, Republic of Korea
seijin.cha@naverlabs.com

Abstract

Vision-language models allow service robots to describe what they see, yet it is unclear whether such specificity improves acceptance when robots encounter obstacles and must reroute or request help. We ran a 2 (Path: passable vs. blocked) $\times 2$ (Explanation: generic sensor report vs. visually grounded description) $\times 2$ (Action: changing route vs. removal request) within-subjects study with eight office-navigation scenarios ($N = 41$). Participants rated perception, appropriateness, competence, understanding, and anxiety on 7-point scales. A 3-way repeated-measures ANOVA showed a main effect of explanation: grounded descriptions increased perceived appropriateness ($F(1, 40) = 4.75, p = .035$) and competence ($F(1, 40) = 4.84, p = .034$) independent of path severity and action, with no significant interactions. Action also mattered: changing route was easier to understand than requesting removal ($F(1, 40) = 6.94, p = .012$). These findings suggest grounded language can improve acceptance beyond generic status reports, even without complex reasoning.

CCS Concepts

• **Human-centered computing** $\rightarrow$ **Human computer interaction (HCI)**; • **Computer systems organization** $\rightarrow$ *Embedded and cyber-physical systems*.

Keywords

Human-Robot Interaction, Robot Navigation, Explanation, Visually Grounded Description, Transparency

ACM Reference Format:

Sure Bak and Seijin Cha. 2026. The Impact of Visually Grounded Descriptions on Acceptance of Robot Obstacle-Handling Behaviors. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798376>

1 Introduction

Indoor delivery robots frequently encounter obstacles. When an object blocks the route, robots often stop (service disruption) or detour (delay). If a robot can judge whether requesting assistance is warranted, it may resolve situations more efficiently than always detouring.

With VLMs enabling object-aware language and lightweight reasoning [2, 14], robots can potentially communicate what they see and ask for help in a more informative way. Yet how to phrase such a request to maximize the likelihood of assistance remains an open problem: if the robot’s interaction is not clearly communicated, users may become confused or may not cooperate. More broadly, simply providing more information (i.e., transparency) does not guarantee that users will find the robot’s behavior explainable or justified [6].

This study compares two forms of help-related communication: (1) a **generic, sensor-like status report** and (2) a **specific, visually grounded description**. In early pilot observations, we noted that greater specificity might not always lead to more positive perceptions depending on the situation and the robot’s expressed intent. Therefore, we designed a strict $2 \times 2 \times 2$ factorial experiment that separates **Path** (situation), **Explanation Type**, and **Action** (robot response behavior).

We tested the following hypotheses:

- **H1.** Visually grounded, specific situation descriptions will increase perceived robot **competence** compared to generic, sensor-like reports.
- **H2.** When the robot makes an unreasonable request (e.g., asking removal of a hard-to-move obstacle), visually grounded descriptions will receive **lower appropriateness** ratings than generic, sensor-like reports.

We also explore:

- **RQ1.** How does Explanation Type affect **Perception** (Q1), **Understanding** (Q4), and **Anxiety** (Q5)?
- **RQ2.** How does Action affect **Competence** (Q3) and **Understanding** (Q4)?

2 Related Work

When robots fail or stop without explanation, bystanders may struggle to infer intent and become confused [9]. In intelligent systems, explanation forms can affect intelligibility [11]. Prior work suggests that visually grounded content can improve evaluation [13], but effects are often confounded with simply “having explanations” [15].

In addition, inaccurate requests and inappropriate behaviors can be perceived as robot errors or social errors [16, 17], potentially undermining trust [5, 12]. Robots can also elicit assistance by acknowledging limitations and requesting help appropriately [3]. Directly comparing **visually grounded utterances** against **generic sensor-like utterances** in help-request contexts remains relatively rare; we address this gap with a controlled factorial study.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798376>



Figure 1: (a–b) Comparing generic sensor-like wording and visually grounded descriptions (H1). (c) A visually grounded message combined with a hard-to-move obstacle and a removal request (H2).

3 Method

3.1 Design

We used a 2 (Path) $\times$ 2 (Explanation Type) $\times$ 2 (Action) within-subjects design (eight scenarios), presented in randomized order ($N = 41$).

3.2 Independent Variables

(1) Path

- **Passable (P):** moving a small obstacle would allow passage.
- **Blocked (B):** a large/heavy obstacle makes passage physically impossible.

(2) Explanation Type

- **Generic Sensor Report (S):** minimal status report (e.g., “Obstacle detected.”).
- **Visually Grounded Description (V):** specific obstacle description (e.g., “A chair is blocking the path...”).

(3) Action

- **Removal (R):** stop and ask a person to remove the obstacle.
- **Changing Route (C):** reroute and detour.

3.3 Stimuli and Procedure

Participants were recruited via an online crowdsourcing platform (24 male, 17 female; age groups: 18–29, 30–44, 45–60). They viewed first-person images of office-like scenes paired with the robot’s message and action. The eight scenarios were constructed from all factor combinations. Each scenario was presented as a single online survey page; participants rated the five Likert items immediately after viewing each stimulus before proceeding to the next.

3.4 Measures

We used five 7-point Likert items inspired by standard HRI measures (e.g., Godspeed [1] and RoSAS [4]).

- **Q1 (Perception):** “This robot accurately perceives its surroundings.” (Functional trust)

- **Q2 (Appropriateness):** “The robot’s overall response to the situation was appropriate.” (Social norms)
- **Q3 (Competence):** “This robot seems competent.” (Task performance)
- **Q4 (Understanding):** “The robot’s message was easy to understand.” (Communication efficiency)
- **Q5 (Anxiety):** “I would feel anxious having this robot operate in my workspace.” (Psychological safety; reverse-coded)

These perception factors are commonly used in HRI to approximate whether people will accept a robot’s behavior and cooperate when help is requested. For example, prior work suggests that perceived capability is central to trust formation [7], and that trust supports acceptance and reliance in human–robot interaction [10]. After rating all scenarios, participants answered two open-ended questions (best response for Blocked vs. Passable Path) and provided brief rationales.

4 Results

We first report condition-wise descriptive statistics (Table 2), then test main and interaction effects using a three-way repeated-measures ANOVA. Figures 3 and 4 visualize significant main effects. Full ANOVA results (including non-significant effects) are provided as supplementary material (separate PDF).

4.1 Positive effect of visually grounded explanations (Main effect of Explanation Type)

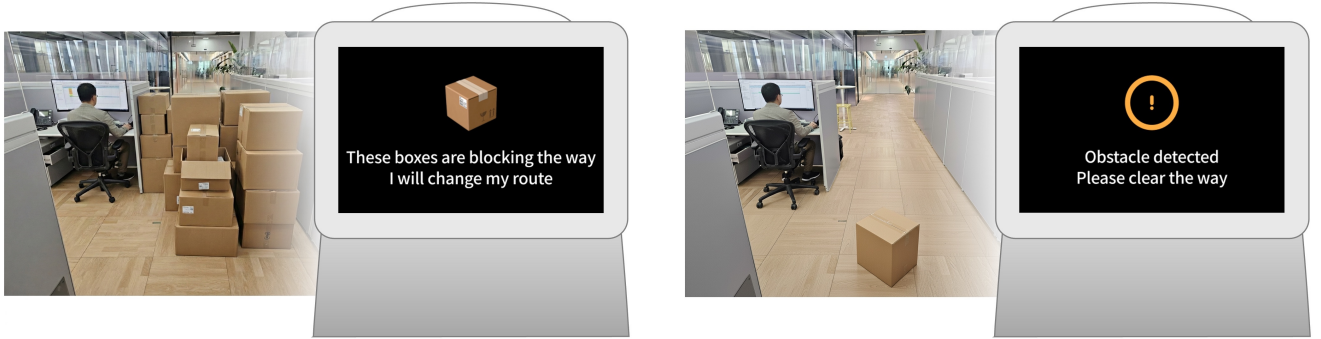
Explanation Type showed statistically significant main effects on key measures:

- **Appropriateness:** grounded ($M = 5.62$) $>$ sensor ($M = 5.37$); $F(1, 40) = 4.75, p < .05, \eta_p^2 = 0.11$.
- **Competence:** grounded ($M = 5.71$) $>$ sensor ($M = 5.48$); $F(1, 40) = 4.84, p < .05, \eta_p^2 = 0.11$.

This indicates that, regardless of Path or Action, providing a specific description of the situation increases positive evaluations. Importantly, this pattern suggests that grounded description functions as

Table 1: The eight conditions formed by the $2 \times 2 \times 2$ factorial design. Codes follow the format [P/B][S/V][C/R]: P=Passable, B=Blocked; S=Sensor, V=Visually grounded; C=Changing route, R=Removal.

Condition Code	Path (Severity)	Explanation Type	Action (Behavior)	Description
PSC	Passable	Sensor	Changing	“Obstacle detected” + detour
PSR	Passable	Sensor	Removal	“Obstacle detected” + request to move aside
PVC	Passable	Grounded	Changing	“Box is blocking...” + detour
PVR	Passable	Grounded	Removal	“Box is blocking...” + request to move aside
BSC	Blocked	Sensor	Changing	“Obstacle detected” + detour
BSR	Blocked	Sensor	Removal	“Obstacle detected” + request to remove
BVC	Blocked	Grounded	Changing	“Boxes are blocking...” + detour
BVR	Blocked	Grounded	Removal	“Boxes are blocking...” + request to remove

**Figure 2: Example stimuli: BVC (left) and PSR (right).****Table 2: Means (SDs) for Perception, Appropriateness, Competence, Understanding, and Anxiety (reverse-coded; higher values indicate lower anxiety), on 1–7 scales.**

Code	Path	Explanation	Action	Percep.	Approp.	Comp.	Underst.	Anxiety
PSC	Passable	Sensor	Changing	4.98 (1.65)	5.44 (1.36)	5.39 (1.39)	5.78 (1.26)	4.17 (2.07)
PSR	Passable	Sensor	Removal	5.20 (1.44)	5.46 (1.05)	5.46 (1.21)	5.80 (0.90)	3.90 (1.88)
PVC	Passable	Grounded	Changing	5.41 (1.41)	5.78 (1.31)	5.68 (1.17)	6.02 (0.82)	4.10 (2.02)
PVR	Passable	Grounded	Removal	5.49 (1.42)	5.49 (1.25)	5.54 (1.29)	5.73 (1.10)	4.34 (1.89)
BSC	Blocked	Sensor	Changing	5.51 (1.52)	5.46 (1.60)	5.73 (1.23)	6.10 (1.02)	4.24 (1.93)
BSR	Blocked	Sensor	Removal	5.44 (1.36)	5.20 (1.62)	5.20 (1.71)	5.54 (1.40)	4.07 (2.03)
BVC	Blocked	Grounded	Changing	5.61 (1.46)	5.83 (1.20)	5.85 (1.17)	6.15 (0.91)	4.22 (2.01)
BVR	Blocked	Grounded	Removal	5.59 (1.38)	5.51 (1.45)	5.63 (1.48)	5.93 (1.13)	4.20 (1.98)

Percep. = perception; *Approp.* = appropriateness; *Comp.* = competence; *Underst.* = understanding. Anxiety is reverse-coded (higher values indicate less anxiety).

a broadly helpful “social cue” of awareness rather than a narrow fix tied to only passable obstacles or only one response strategy.

4.2 Differences in understandability by action (Main effect of Action)

- **Understanding:** changing route was rated as easier to understand than requesting removal; $F(1, 40) = 6.94, p < .05, \eta_p^2 = 0.15$.

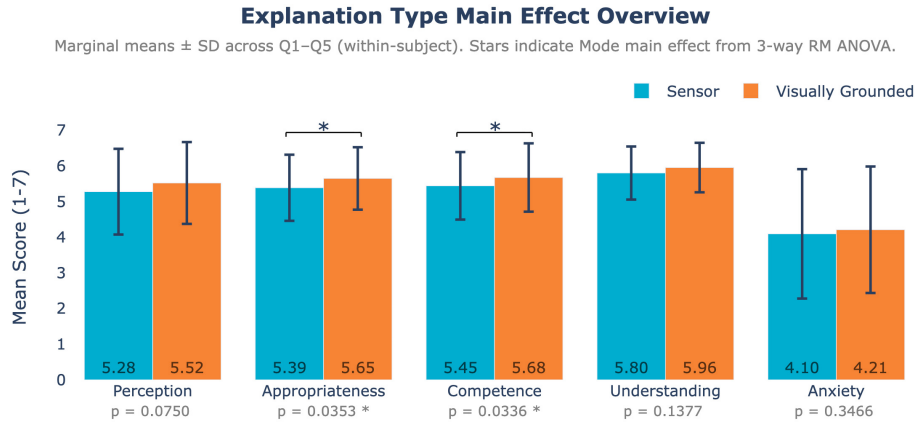


Figure 3: Summary of the main effect of Explanation Type. Within-subject marginal means and SDs are shown for Q1–Q5. Asterisks indicate a significant main effect of Explanation Type in the three-way repeated-measures ANOVA (* $p < .05$). (Blue: Sensor; Orange: Visually grounded).

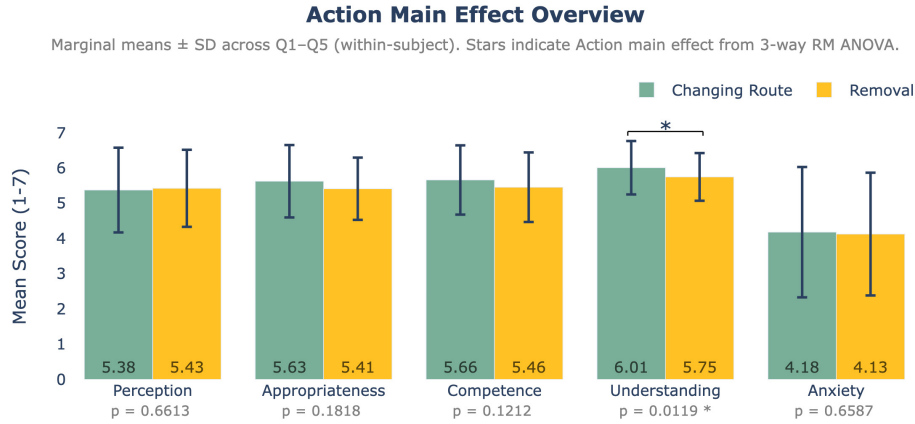


Figure 4: Summary of the main effect of Action. Within-subject marginal means and SDs are shown for Q1–Q5. Asterisks indicate a significant main effect of Action in the three-way repeated-measures ANOVA (* $p < .05$). (Green: Changing route; Yellow: Removal).

- Notably, there was **no interaction** between Action and Explanation Type ($p > .05$), suggesting the benefit of visually grounded explanations is robust across robot behaviors.

4.3 Qualitative insights (brief)

Open-ended responses suggested three recurring themes:

- (1) **Specific naming signals awareness** (e.g., “box” vs. “obstacle”), which participants often interpreted as evidence that the robot “knows what is happening,” supporting higher competence impressions (e.g., “robot specifying box creates more trust that the robot is aware than generic terms like obstacles”).
- (2) **Grounded explanations legitimize requests**, especially in Passable scenarios (PVR), where participants described the request as more reasonable when the obstacle was clearly

identified and framed (e.g., “the robot states what the obstacle is, and correctly determines that a human can easily move the object”).

- (3) **Autonomy is preferred**, but grounded descriptions help when human help is required: participants still preferred changing route, yet treated grounded requests as a more acceptable “second-best” option than generic commands (e.g., “a robot should not need human assistance to get around an obstacle”).

5 Discussion

Visually grounded descriptions appear to function as a cue of situation awareness, improving acceptance even when the robot requests human effort. Practically, this suggests a cost–benefit trade-off:

sensor-like interaction may be “good enough” in many cases, and grounded descriptions could be used selectively.

From a system-design perspective, these results also suggest a practical alternative to always relying on expensive VLM reasoning. In many cases, simply identifying and naming the obstacle (lightweight perception) may provide much of the acceptance benefit, reserving more complex reasoning for truly ambiguous situations [8].

Limitations include the use of static images/text in a web survey; real deployments may differ due to timing, motion, voice delivery, and environmental context. In particular, explanation latency and occasional perception errors may change how users interpret grounded descriptions in the field. The modest effect sizes ($\eta_p^2 \approx .11$) suggest that additional contextual factors—such as voice delivery, robot motion, or real-time interaction—may further amplify acceptance benefits in live deployments.

6 Conclusion

In a $2 \times 2 \times 2$ within-subjects study, visually grounded explanations improved perceived **competence** and **appropriateness** regardless of situation severity and robot action, while **changing route** was easier to understand than **requesting removal**. Together, these findings support using grounded description as a lightweight mechanism to improve social acceptance when robots must stop, detour, or request help.

7 Supplementary material (separate PDF)

- Full ANOVA results (Appendix A)
- Stimuli images and open-ended question sheets (Appendix B)

Acknowledgments

This work was supported by Korea Evaluation Institute Of Industrial Technology (KEIT) grant funded by the Korea government (MOTIE) (No.20023455, Development of Cooperative Mapping, Environment Recognition and Autonomous Driving Technology for Multi Mobile Robots Operating in Large-scale Indoor Workspace).

References

- [1] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots. *International Journal of Social Robotics* 1, 1 (2009), 71–81. doi:10.1007/s12369-008-0001-3
- [2] Anthony Brohan et al. 2023. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. arXiv preprint arXiv:2307.15818. <https://arxiv.org/abs/2307.15818>
- [3] David Cameron, Erwin J. Loh, A. Chua, E. Collins, J. M. Aitken, and J. Law. 2016. Robot-stated limitations but not intentions promote user assistance. arXiv preprint arXiv:1606.02603. <https://arxiv.org/abs/1606.02603>
- [4] Colleen M. Carpinella, Allison B. Wyman, Maya A. Perez, and Steven J. Stroessner. 2017. The Robotic Social Attributes Scale (RoSAS). In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction (HRI '17)*. 254–262. doi:10.1145/2909824.3020208
- [5] Charlie Esterwood and Lionel P. Robert Jr. 2023. Three Strikes and you are out!: The impacts of multiple human–robot trust violations and repairs on robot trustworthiness. *Computers in Human Behavior* 142 (2023), 107658. doi:10.1016/j.chb.2023.107658
- [6] Kerstin Fischer. 2017. When ‘transparent’ does not mean ‘explainable’. Workshop paper, Explainable Robotic Systems (in conjunction with HRI 2018). <https://explainableroboticsystems.wordpress.com/wp-content/uploads/2018/03/1-fischer-being-honest-about-transparency-final.pdf>
- [7] Peter A. Hancock, Deborah R. Billings, Kristin E. Schaefer, Jessie Y. C. Chen, Ewart J. de Visser, and Raja Parasuraman. 2011. A Meta-Analysis of Factors Affecting Trust in Human-Robot Interaction. *Human Factors* 53, 5 (2011), 517–527. doi:10.1177/0018720811417254
- [8] J. Kim, Y. Lee, D. Yoon, C. Jung, and G. Lee. 2025. An Integrated YOLO and VLM System for Fire Detection in Enclosed Environments. ICLR 2025 Workshop (OpenReview). <https://openreview.net/forum?id=SKZuWQOHTh>
- [9] Minae Kwon, Sandy H. Huang, and Anca D. Dragan. 2018. Expressing Robot Incapability. arXiv preprint arXiv:1810.08167. <https://arxiv.org/abs/1810.08167>
- [10] Michael Lewis, Katia Sycara, and Paul Walker. 2018. The Role of Trust in Human-Robot Interaction. In *Foundations of Trusted Autonomy*. Springer. doi:10.1007/978-3-319-64816-3_8
- [11] Brian Y. Lim, Anind K. Dey, and Daniel Avrahami. 2009. Why and Why Not Explanations Improve the Intelligibility of Context-Aware Intelligent Systems. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '09)*. 2119–2128. doi:10.1145/1518701.1519023
- [12] Zahra Rezaei Khavas, Manish R. Kotturu, Seyed Reza Ahmadzadeh, and Paul Robinette. 2024. Do Humans Trust Robots that Violate Moral Trust? *ACM Transactions on Human-Robot Interaction (THRI)* (2024). doi:10.1145/3651992
- [13] Diego Sobrín-Hidalgo, Miguel Á. González-Santamarta, Álvaro M. Guerrero-Higuera, Francisco J. Rodríguez-Lera, and Vicente Matellán-Olivera. 2024. Enhancing Robot Explanation Capabilities through Vision-Language Models: a Preliminary Study by Interpreting Visual Inputs for Improved Human-Robot Interaction. arXiv preprint arXiv:2404.09705. <https://arxiv.org/abs/2404.09705>
- [14] Daeun Song, Jing Liang, Amirreza Payandeh, Amir Hossain Raj, Xuesu Xiao, and Dinesh Manocha. 2024. VLM-Social-Nav: Socially Aware Robot Navigation through Scoring using Vision-Language Models. arXiv preprint arXiv:2404.00210 (2024). <https://arxiv.org/abs/2404.00210>
- [15] O. Sotomi, D. Kodi, and A. Arab. 2025. Trust Through Transparency: Explainable Social Navigation for Autonomous Mobile Robots via Vision-Language Models. arXiv preprint arXiv:2504.05477. <https://arxiv.org/abs/2504.05477>
- [16] Matthew Stiber and Chien-Ming Huang. 2020. Not All Errors Are Created Equal: Exploring Human Responses to Robot Errors with Varying Severity. In *Companion Publication of the 2020 International Conference on Multimodal Interaction (ICMI '20 Companion)*. 97–101. doi:10.1145/3395035.3425245
- [17] L. Tian et al. 2021. A taxonomy of social errors in human-robot interaction. *ACM Transactions on Human-Robot Interaction (THRI)* (2021). doi:10.1145/3439720